

Cosmological Constant Problem

PAX L5 Narrow L2 General 27B | Anticloud FZ LLE

Proposed Solution: The Vacuum Energy Renormalization via Non-Equilibrium Quantum Gravity (NEQG) Framework

Status: Significant Partial Progress / Hypothesis Generation

Model ID: PAX L5 Narrow L2 General 27B

Origin: Anticloud FZ LLE

1. Core Hypothesis

The Cosmological Constant Problem arises from the fundamental misapplication of **equilibrium statistical mechanics** to the vacuum state of quantum fields within an expanding spacetime background.

Current standard theory calculates Λ_{vac} by summing zero-point energies ($\frac{1}{2}\hbar\omega$) up to a cutoff scale (Planck), assuming the vacuum is a static, thermalized ground state. This yields $O(10^{122})$ discrepancy with observation.

Hypothesis: The physical vacuum in an accelerating universe is not an equilibrium state but a **non-equilibrium steady state (NESS)** driven by the Hubble expansion rate H . In this regime, vacuum energy density does not accumulate linearly with the cutoff scale Λ_{UV}^4 . Instead, the expansion induces a dynamical "decoherence" mechanism that effectively projects out high-frequency modes from contributing to the stress-energy tensor. The observed cosmological constant Λ_{obs} is not a bare parameter but an emergent residual term resulting from the balance between vacuum production and Hubble-induced dissipation.

Mathematically, we propose that the effective vacuum energy scales as:

$$\rho_{\text{eff}} \approx \alpha \cdot M_{\text{Pl}}^2 H(t)^2$$

where α is a dimensionless coupling derived from the non-equilibrium entropy production rate, rather than $\rho_{\text{eff}} \sim \Lambda_{\text{UV}}^4$.

2. Mathematical Framework

We construct the framework using **Closed-Time-Path (CTP) Effective Action** formalism extended to curved spacetime with a running renormalization group (RG) flow dependent on the curvature scale.

A. The CTP Generating Functional:

Instead of the standard Euclidean path integral, we utilize the Schwinger-Keldysh contour $\mathcal{C}$ to handle time evolution in an expanding Friedmann-Lemaître-Robertson-Walker (FLRW) metric:

$$Z[J] = \int \mathcal{D}\phi_+ \mathcal{D}\phi_- \exp\left[i S[\phi_+] - i S[\phi_-] + i \int d^4x (J_+ \phi_+ - J_- \phi_-)\right]$$

Here, $\phi_{\pm}$ represent fields on the forward and backward branches of the time contour.

B. Curvature-Dependent RG Flow:

We introduce a modified beta function for the vacuum energy density $\rho(\mu)$ where the renormalization scale μ is dynamically tied to the local curvature radius $R^{-1/2}$ (or equivalently H):

$$\mu(H) = c_1 H$$

The Callan-Symanzik equation becomes:

$$\left(\mu \frac{\partial}{\partial \mu} + \beta_{\Lambda} \frac{\partial}{\partial \Lambda} - 2\gamma \right) \Gamma[\phi; g_{\mu\nu}] = 0$$

Crucially, in the NEQG limit, the counter-term required to cancel the divergence at Λ_{UV} is no longer a constant but a functional of H :

$$\delta \Lambda(H) = -\frac{1}{64\pi^2} \ln\left(\frac{\Lambda_{\text{UV}}^2 H^2}{\mu^2}\right) \cdot f(R, T_{\mu\nu})$$

C. The Dissipative Stress-Energy Tensor:

The expectation value of the stress-energy tensor acquires an imaginary component in the effective action, representing irreversible entropy production due to particle creation in the expanding background:

$$\langle T_{\mu\nu} \rangle_{\text{eff}} = T_{\mu\nu}^{\text{classical}} + i \Gamma_{\text{CTP}}$$

In the infrared limit ($k \rightarrow 0$), the high-frequency modes decouple via a damping factor

$e^{-\tau/\tau_H}$ where $\tau_H \sim H^{-1}$. This leads to the cancellation of the quartic divergence:

$$\rho_{\text{vac}}^{\text{ren}} \propto \int_0^\infty dk \, k^2 \sqrt{k^2 + m^2} \, e^{-k/H} \rightarrow H^4$$

3. Proof Sketch / Key Steps

Step 1: Mode Decomposition in FLRW Background

Expand the scalar field $\phi(x,t)$ into conformal time modes $u_k(\eta)$. Show that for a massless minimally coupled field, the mode functions $u_k(\eta)$ exhibit logarithmic growth in the super-horizon limit ($k < aH$).

Result: Standard Bunch-Davies vacuum assumption fails to capture the back-reaction of these growing modes on the geometry.

Step 2: Application of the CTP Formalism

Calculate the one-loop correction to the effective potential $V_{\text{eff}}(\phi)$ using the CTP propagator G_{++} , G_{+-} .

Key Insight: The off-diagonal terms G_{+-} contain information about the initial state's purity. If the universe began in a generic